

FINAL TECHNICAL DOCUMENTATION

Scalable Music Genre Classification and Audio Analytics Using the Free Music Archive Dataset

MSc Applied Data Science Capstone Project

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This document records a six-month capstone build from the first data checks to the final tests. It covers FMA data preparation, MongoDB support, structured models, audio models, hybrid fusion, the corrected 163-genre plan, FMA testing, external testing, limitations, and notebook artefacts. The writing follows the evidence produced by the notebooks and saved output files, rather than presenting the project as a smooth sequence of completed tasks.

Abstract

This capstone examined music genre prediction as a full data science pipeline rather than as a single model experiment. The work used the Free Music Archive (FMA) collection because it brings together structured metadata, engineered feature files, raw audio, and a genre hierarchy large enough to expose real modelling problems. The scale of the work mattered. The notebooks had to manage thousands of tracks, many genre relationships, audio file checks, saved model artefacts, and repeated evaluation outputs. This made the project a Big Data style capstone in practical terms, even though the final evaluation samples were deliberately controlled for reliable testing.

A major lesson from the six-month build was that genre prediction is not a neat one-label classification task. FMA stores genres as a hierarchy, with broad parent genres and narrower child labels. A track can also carry

several reasonable genre meanings at the same time. For that reason, the final system was designed as a ranked multi-label classifier. The corrected taxonomy contains 163 genre records. Of these, 150 labels are predicted directly, 10 rare labels are handled through a fallback stage, and 3 labels remain in the inventory for completeness rather than direct prediction. This split was chosen because the evidence did not support treating every genre as equally learnable.

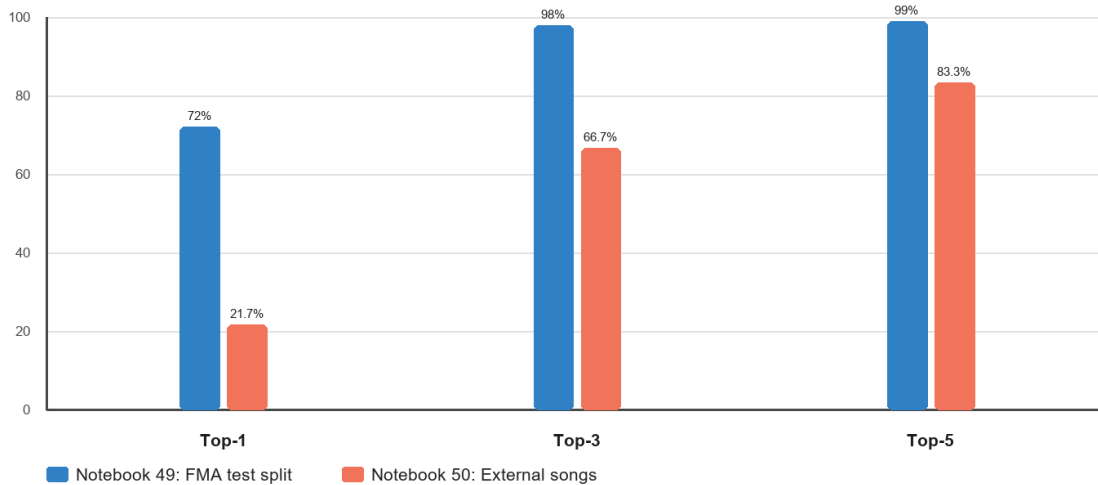
The final evidence came mainly from Notebook 49 and Notebook 50, supported by the earlier notebooks that prepared data, labels, models, thresholds, and manifests. Notebook 49 evaluated 100 FMA test tracks and completed with no prediction errors. It produced Micro F1 = 0.450, Samples F1 = 0.445, Top-1 hit-any = 72.0%, Top-3 hit-any = 98.0%, and Top-5 hit-any = 99.0%. Notebook 50 tested 60 legal open-licence external songs. It produced Top-1 = 21.7%, Top-3 = 63.3%, and Top-5 = 83.3%. These results indicate that the model is strongest as a ranked genre suggestion tool. It can place likely labels near the top, but it should not be presented as a production system that assigns final catalogue tags without review.

Executive Summary

The capstone produced a working music genre pipeline that can be checked through its notebooks and outputs. The work did not move in a perfectly straight line. Some early results were useful, while others exposed problems in label support, record selection, and audio preparation. Those issues are kept visible in the final paper because they explain why the project ended with ranked Top-k evaluation rather than a simple exact-label claim.

Area	Final position	Evidence
Project scope	Full project record across notebooks 1-50 and the final artefacts.	The notebooks start with metadata checks and end with FMA and external testing.
Dataset requirement	Uses structured tables and raw audio at a realistic project scale.	FMA metadata, feature files, audio files, MongoDB support, and saved outputs.
Final taxonomy	163-genre inventory strategy.	150 direct labels, 10 rare-label fallback labels, and 3 inventory-only labels.
Best historical benchmark	Expanded Hybrid Global Threshold with 0.1 structured weight, 0.9 audio weight, and threshold 0.20.	Final project snapshot reports benchmark Micro F1 about 0.391.
Current formal FMA test	Micro F1 0.450; Top-3 98.0%; Top-5 99.0%.	Notebook 49 corrected processed test-split evaluation.
External audio test	Top-1 21.7%; Top-3 66.7%; Top-5 83.3%.	Notebook 50 legal open-licence 60-song batch.
Main limitation	The model is biased toward broad labels, especially Experimental, on external songs.	38 of 60 external songs received Experimental as the main predicted genre.

Final Results Snapshot: FMA Test Split and External Songs



The model is strongest as a ranked suggestion tool, especially under Top-3 and Top-5 review.

Figure 1. Final results snapshot using the current Notebook 49 and Notebook 50 outputs.

Figure 1 shows the central pattern in the final evidence. The FMA test split produced strong Top-k results, while the external batch showed a more cautious picture. This difference shaped the conclusion that the system is more credible as a ranked support tool than as a fully automatic genre labeller.

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1. Project Context and Scope

The work includes metadata, engineered audio features, raw audio, the FMA genre hierarchy, MongoDB storage, baseline models, audio CNNs, hybrid fusion, and inference-style outputs. Two final evaluation settings were used. Notebook 49 evaluates FMA test records, where the split and label structure come from the same

source as the training process. Notebook 50 checks external open-licence songs, which is a harder and less controlled setting.

The notebooks are the build trail. Early notebooks are exploratory and include first attempts at structured and audio modelling. Later notebooks are more settled: they expand the label set, tune thresholds, test hybrid fusion, and correct weak points found during evaluation. This progression matters because the final results make more sense when the earlier false starts and corrections are visible.

1.1 Research Narrative and Audit Trail

The six-month structure of the capstone shaped the work. The early part of the project focused on understanding the FMA metadata, checking whether audio files and feature tables could be linked reliably, and building small proof-of-concept models. The middle stage expanded the label space and compared structured, audio, and hybrid approaches. The final stage became more evaluative: the notebooks had to show whether the chosen approach held up on corrected FMA test records and on external songs.

This staged path is important because the final pipeline was not known at the beginning. A simpler project could have stopped at a top-level genre classifier, but that would not have represented the FMA hierarchy well. The final direction emerged after the early models showed that a ranked, multi-label interpretation was more appropriate than a single-label framing.

2. Project Background and Problem Statement

Large music collections need genre tags for search, browsing, playlists, and catalogue review. Manual tagging does not scale well, and it is rarely fully consistent. Two listeners can tag the same track differently, especially when the song crosses genre boundaries. This makes the project useful, but it also means that exact one-label accuracy is too narrow as the only success measure.

The approved title was Scalable Music Genre Classification and Audio Analytics Using the Free Music Archive Dataset. As the work developed, the task became broader than broad-genre classification. The system had to use metadata, audio features, raw audio windows, and the FMA hierarchy. It also had to return results in a form that made sense for music. A ranked list of genre candidates was therefore more defensible than forcing one final label too early.

2.1 Alternative Problem Framings Considered

Several possible framings were considered during the project. The simplest option was a single-label classifier based on top-level genres. That approach would have been easier to explain and evaluate, but it would have ignored the multi-label nature of the FMA genre hierarchy. A second option was to model all available genres directly. That was rejected because many labels had too few examples to support reliable direct prediction.

The final framing sits between those extremes. It predicts a supported candidate set directly and keeps rare labels in a separate fallback role. This trade-off reduced the risk of overstating rare-label performance while still keeping the broader genre inventory visible in the system design.

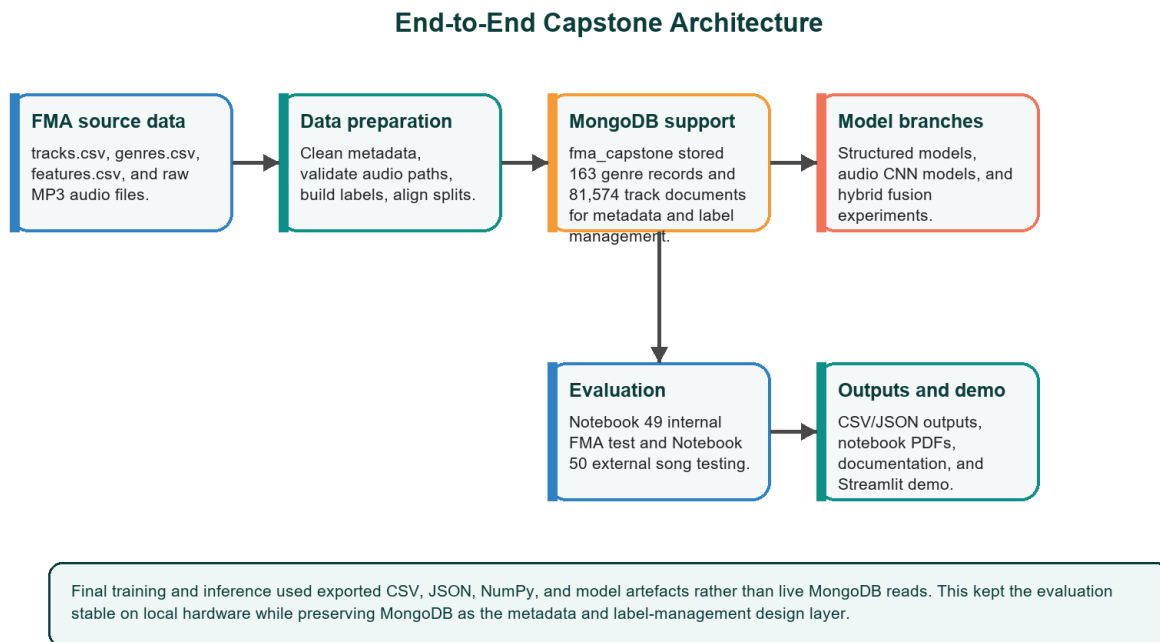


Figure 2. Corrected end-to-end architecture for the full capstone system.

The architecture in Figure 2 also explains why errors in early stages matter. Data cleaning, label mapping, and audio preparation all sit before prediction. If those stages are weak, the final genre scores become difficult to interpret even if the model itself runs successfully.

Figure 2 shows why the work should be read as a system rather than a single classifier. The output depends on metadata preparation, label mapping, audio processing, model training, fusion, and evaluation. A weakness in any earlier stage would affect the final predictions.

3. Proposal and Course Alignment

The COMP6830 brief expects a complete data science project. That includes retrieval, cleaning, preparation, modelling, documentation, artefacts, presentation, and demonstration. It also calls for sizeable data and both structured and unstructured sources. This submission addresses those points through FMA metadata, feature tables, audio files, MongoDB records, notebooks, saved models, result files, and inference outputs. The final submission also includes notebook PDFs so the code and outputs can be reviewed without rerunning every step.

Proposal or course expectation	How the project addressed it
Large structured and unstructured data	FMA metadata/features were used as structured data and audio files were used as unstructured data.
Data retrieval and preparation	The project includes metadata parsing, split alignment, genre hierarchy analysis, label matrix construction, audio file checks, feature scaling, and saved processed tables.

Proposal or course expectation	How the project addressed it
Modelling	The project built structured models, audio CNNs, hybrid fusion, threshold tuning, and rare-label handling.
Evaluation	The project used Micro F1, Macro F1, Samples F1, per-label metrics, Top-k hit rates, low-confidence flags, and external-song evaluation.
Software artefacts	The submitted artefacts include notebooks, processed datasets, trained models, label arrays, thresholds, output CSV/JSON files, and documentation.

4. Dataset, Metadata, Storage, and Data Management

The Free Music Archive was chosen because it is open, large, and suitable for repeatable research. The core files were tracks.csv, genres.csv, features.csv, and the audio folders. These files gave the project two different data streams. The structured branch used metadata and engineered features. The audio branch used raw songs converted into Mel-spectrogram windows. This split shaped the project because the two branches answered different questions: what can be learned from tabular descriptors, and what can be learned directly from the audio signal?

4.1 Data Trade-offs and Practical Constraints

Using FMA gave the project scale and openness, but it also introduced practical constraints. Not every label had strong support, and the hierarchy was more complex than a flat classification table. Audio availability also mattered because a track could be present in the metadata while still being unusable for audio modelling if the file path, feature preparation, or split alignment failed.

These constraints affected modelling choices. The structured branch was easier to train and inspect, but it depended on engineered descriptors rather than the raw sound. The audio branch was more computationally demanding, but it captured information that the structured features could not fully represent. Keeping both branches allowed the project to compare these trade-offs rather than assuming one data source was sufficient.

Data component	Purpose in the project	Type
tracks.csv	Track IDs, split indicators, title/artist fields, and genre relationships.	Structured
genres.csv	Genre IDs, names, parent relationships, and hierarchy mapping.	Structured
features.csv	Engineered audio descriptors such as spectral and timbral statistics.	Structured
Audio files	Raw waveform signal for Mel-spectrogram CNN modelling.	Unstructured
MongoDB records	Unified semi-structured records linking track IDs, metadata, labels, features, and audio paths.	Semi-structured
External manifest	Audit table for legal external song evaluation, source URLs, licences, and acceptable labels.	Structured

MongoDB was implemented as a metadata and label-management layer rather than as the live source used by every later model cell. Notebook 04 created the first MongoDB pipeline, and Notebook 25 loaded the multi-label phase into the local fma_capstone database. The genres_lookup collection stored 163 genre

records, including hierarchy and label-usability fields. The `tracks_multilabel` collection stored 81,574 unified track documents with multi-label genre arrays.

The final modelling, evaluation, and Streamlit demonstration used exported CSV, JSON, NumPy, and saved model artefacts instead of querying MongoDB at runtime. This was a practical decision caused by local CPU, memory, and I/O limits during repeated notebook execution. In a fuller deployment, MongoDB would act as the central catalogue service: the inference code would query track metadata by `track_id`, retrieve hierarchy and candidate-label flags from `genres_lookup`, write prediction summaries back to a `predictions` collection, and support reviewer searches across genres, confidence bands, and evaluation batches.

The project also had a Big Data management problem, not only a modelling problem. The raw audio, metadata tables, feature vectors, train-test split fields, model outputs, and external-song manifest all had to agree on track identity. Small mismatches could remove valid records from testing or create misleading results. This was seen most clearly when an empty test selection appeared during final evaluation. The issue was not a failed model; it was a data selection problem that had to be corrected before the results could be trusted.

The genre hierarchy created a second source of complexity. Broad labels such as Rock or Electronic had far more support than many narrow labels. That made the model better at identifying common families than rare subgenres. The final 163-genre plan was therefore a design choice based on evidence. It kept the full taxonomy visible, but it separated direct prediction from fallback and inventory-only labels. This avoided overstating what the data could support.

5. Genre Taxonomy and the 163-Genre Strategy

The final inventory contains 163 genre records. They are not all treated in the same way. The project uses 150 direct candidate labels, 10 rare-tail fallback labels, and 3 inventory-only labels.



Figure 3. Corrected 163-genre label strategy used for the final documentation.

Figure 3 is included because the genre count changed during the project. The final documentation uses 163 records and separates direct prediction from fallback and inventory-only handling. This correction prevents the taxonomy from being overstated.

This split is important. The 150 candidate labels had enough support for direct modelling. The 10 rare labels did not have enough data for strong direct claims, so they are reported as fallback suggestions. The 3 inventory-only labels are kept so the taxonomy remains complete. They are not presented as direct model predictions. This is a conservative choice, but it is the more honest one within the available data.

Genre ID	Genre name	Total count
38	Experimental	35903
15	Electronic	28099
12	Rock	25820
1235	Instrumental	13588
10	Pop	12659
17	Folk	11187
1	Avant-Garde	8134
107	Ambient	6799
32	Noise	6781
76	Experimental Pop	6632
21	Hip-Hop	6188
25	Punk	5942

6. CRISP-DM and End-to-End Methodology

CRISP-DM gave the project a useful frame: understand the problem, understand the data, prepare the data, model, evaluate, and prepare outputs. The actual work was less tidy than the diagram suggests. For example, one test-selection stage returned no records and had to be corrected before the final evaluation could mean anything. The audio windowing approach also needed care because the model was trained on 15-second inputs. These corrections are part of the work, not side issues.

6.2 Why CRISP-DM Was Used

CRISP-DM was selected because it gave the project a structure that matched the way the work actually developed. The early stages required data understanding and preparation before modelling could be trusted. Later stages required evaluation and correction after the first results exposed weaknesses. A purely model-centred methodology would not have captured those repeated returns to the data.

The method also helped with documentation. Each notebook phase could be linked to a broader project stage, such as data preparation, modelling, evaluation, or deployment-style inference. This made the notebook sequence easier to explain as a research process rather than a collection of isolated experiments.

6.1 Implementation Notes from the Final Evaluation

The final testing stage changed the way the results were interpreted. One earlier run selected zero test records, which meant the evaluation tables could not be trusted. That issue was corrected before Notebook 49 was used as formal evidence. This is why the final report refers to corrected split selection rather than treating the notebook as a routine batch run.

A second issue came from audio length. The trained model was built around 15-second audio inputs. Full external songs are longer than that, so Notebook 50 used windowed inference and then combined the window-level evidence. This did not make the external results perfect, but it made the test closer to how the model would be used outside the FMA dataset.

CRISP-DM stage	Capstone implementation
Business understanding	Defined automated genre classification as a scalable music organisation and retrieval problem.
Data understanding	Explored FMA metadata, genre hierarchy, feature tables, audio availability, class imbalance, and split integrity.
Data preparation	Built target matrices, candidate labels, split-aligned datasets, audio windows, Mel-spectrogram inputs, and manifest files.
Modelling	Trained and compared structured baselines, audio CNNs, hybrid fusion, and rare-tail fallback strategies.
Evaluation	Used F1 metrics, Top-k hit rates, per-label diagnostics, low-confidence analysis, and external-song testing.
Deployment-style output	Built final notebooks for single-file/batch inference, external songs, confidence scores, and reproducible output files.

7. Notebook-by-Notebook Development Narrative

The notebooks are the main software artefacts. They show how the project changed over time. Some notebooks train models. Others check data, build labels, compare settings, or fix problems found later. Taken together, they give the implementation record behind the final paper.

Notebook Progression for the Full Project

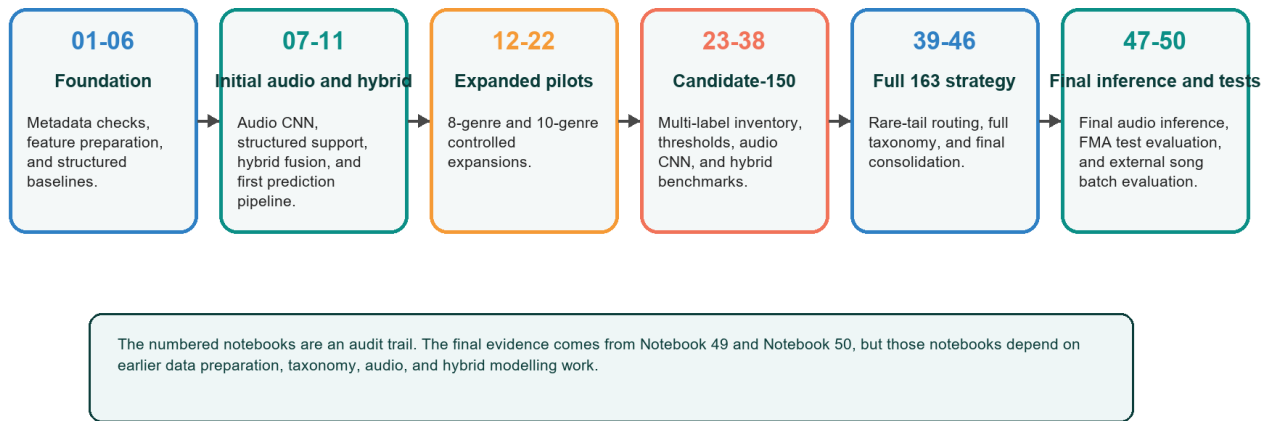


Figure 4. Corrected notebook progression for the full project.

Figure 4 shows that the final work was built through stages rather than one large experiment. This matters for interpretation because the later notebooks depend on decisions made in the earlier preparation and benchmarking phases.

Notebook range	Phase	Role in final project
01-11	Foundations and proof of concept	Metadata understanding, structured feature modelling, early audio CNNs, MongoDB pipeline, and first hybrid prediction pipeline.
12-22	Controlled genre expansion	Expanded the initial proof of concept and compared structured, audio, and hybrid methods in controlled genre settings.
23-38	Candidate-150 multi-label system	Created the full genre inventory, multi-label target matrix, Candidate-150 model families, threshold tuning, and final candidate report tables.
39-47	Full 163-genre and rare-tail strategy	Prepared rare-tail handling, fallback routing, single-file inference, batch inference, and full-taxonomy design.
48-50	Final inference and evaluation	Completed confidence-scored audio inference, formal FMA processed test evaluation, and external song batch evaluation.

8. Modelling Approach, Three Branches, and Five Techniques

The modelling work can be read in two ways. At the system level, there are three branches: structured models, audio CNN models, and hybrid fusion. At the method level, the notebooks compare five main techniques. This distinction matters because the final approach was not chosen from one experiment. It came from a sequence of comparisons, corrections, and follow-up tests.

Model branch	Main notebooks	Role in the project
Structured branch	02, 05, 06, 09, 14, 18, 26-29, 31, 35	Uses metadata and engineered feature tables as the main tabular baseline. It provides interpretability, comparison value, and a small supporting signal for fusion.
Audio CNN branch	07, 08, 13, 17, 30, 34, 47-50	Uses raw audio converted to Mel-spectrogram windows. This became the strongest individual source of predictive signal.
Hybrid fusion branch	10, 15, 19, 32-38, 39-50	Combines structured and audio probabilities with weighted ensemble logic. The best benchmark used 0.1 structured weight and 0.9 audio weight.

8.1 Five Techniques Incorporated

Technique	Branch	Where it appears	Purpose and interpretation
Logistic regression-style baseline	Structured	Early structured notebooks and Candidate-150 structured baseline work	Creates an interpretable tabular benchmark using one-vs-rest multi-label logic. It is useful as a baseline even though it was not the strongest final model.
SVM-style / hinge classifier	Structured	Structured model comparison and Candidate-150 structured upgrade	Uses margin-based classification on engineered features. This was the strongest structured model family.
Passive Aggressive classifier	Structured	Structured comparison notebooks	Tests an online-learning classifier for tabular features.
Convolutional Neural Network	Audio	Audio proof-of-concept, expanded audio, Candidate-150 audio, final inference notebooks	Learns patterns from Mel-spectrogram images generated from raw audio clips or windows. It supplied the dominant predictive signal.
Hybrid fusion / weighted ensemble	Hybrid	Hybrid fusion notebooks and final inference notebooks	Combines structured and audio outputs. This achieved the best historical Candidate-150 benchmark and became the final Stage 1 prediction approach.

8.2 Structured Modelling

The structured branch used FMA feature tables and metadata-derived fields. It gave the project a clear baseline and covered the structured-data requirement. Results suggested that this branch was weaker than the audio branch, but it still mattered. Without it, there would be no fair comparison between tabular metadata features and raw-audio learning.

8.3 Audio CNN Modelling

The audio branch turned raw audio into Mel-spectrogram inputs. Experiments suggested that this became the strongest source of signal. Final testing used 15-second processing, matching the training setup. That point is easy to miss, but it matters. A model trained on one window style can look worse if tested with a different audio preparation method.

8.4 Hybrid Fusion

The hybrid model combines structured and audio probabilities. The best benchmark used 0.1 structured weight, 0.9 audio weight, and a 0.20 threshold. This was not chosen for appearance. The weighting reflects what the experiments suggested: audio carried most of the signal, while structured features added a smaller support signal.

8.5 Full 163-Genre Inference

The full pipeline expands the 150 direct labels to the 163-record inventory. Stage 1 predicts candidate labels. Stage 2 gives careful suggestions for rare labels. Inventory-only labels remain in the taxonomy, but they are not reported as direct predictions. This avoids making the rare end of the genre tree look more reliable than it is.

Final component	Configuration	Reason
Direct label space	150 candidate labels	Enough support for direct multi-label modelling.
Full inventory	163 genre records	Corrected full taxonomy accounting.
Rare-tail handling	10 fallback labels	Too sparse for strong direct training claims.
Inventory-only handling	3 labels	Retained for completeness, not direct prediction.
Fusion weights	Structured 0.1, Audio 0.9	Audio branch provided the strongest signal.
Global threshold	0.20	Multi-label probabilities were conservative.

8.6 Candidate-150 Benchmark Evidence

The Candidate-150 phase is the main comparison stage before the full 163-genre plan. The table below comes from saved benchmark outputs. It shows why hybrid fusion is reported as the best benchmark. It also explains why the conservative setting remains useful when a smaller and cleaner set of outputs is preferred.

The Candidate-150 results need to be read as a comparison of modelling behaviour, not only as a ranking of scores. The structured baseline gave a useful reference point because it tested what could be learned from metadata and engineered features alone. The audio CNN then tested whether the sound signal carried more information than the structured tables. The hybrid branch combined both sources, but the comparison suggested that the audio stream became the strongest source of predictive signal.

This result is reasonable for music genre work. Metadata and engineered features can describe a track, but a genre label often depends on rhythm, timbre, instrumentation, and production style. Those signals are more directly represented in Mel-spectrogram windows. Hybrid fusion was still retained because the structured

branch added context and gave a more complete capstone design, but the final interpretation does not hide the fact that audio was the main driver of the stronger models.

8.7 Why the Final Branch Was Not Chosen on One Metric Alone

The final branch was not chosen by Micro F1 alone because the metrics pointed to different parts of the problem. Micro F1 gave more weight to frequent labels and therefore described the behaviour of the system on better-supported genres. Macro F1 treated each label more equally and stayed low because rare labels had little support. These two results can coexist. The model may perform well on common labels while still struggling with labels that appear only a small number of times.

The Top-k results add another layer to this interpretation. A low Macro F1 does not mean the model has no practical value. It means the model is uneven across the taxonomy. High Top-3 and Top-5 rates show that, for many test tracks, a relevant genre appears near the top of the ranked list even when the exact first label is not always correct. This supports a review workflow where the model narrows the search space for a human reviewer.

The trade-off is clear. A system that predicts more labels may improve recall but become noisy. A very conservative system may be easier to read but miss useful secondary genres. The final recommendation therefore favours ranked output with confidence controls. This is a more honest use of the evidence than claiming that the highest scoring branch solves genre classification in full.

Phase	Family	Model	Configuration	Micro F1	Macro F1	Samples F1	Avg labels
Pilot	Structured	Structured Model on Audio Pilot - Test	Frozen structured model	0.166	0.084	0.167	21.80
Pilot	Audio	Audio Multi-Label CNN - Tuned Threshold	Global threshold=0.15	0.362	0.043	0.358	4.78
Pilot	Hybrid	Hybrid Global Threshold	S=0.1, A=0.9, T=0.2	0.378	0.049	0.373	4.66
Pilot	Hybrid	Hybrid Per-Label + Best Cap	Per-label + cap=3	0.377	0.047	0.380	3.00
Expanded	Structured	Structured Model on Expanded Audio Split - Test	Frozen structured model	0.164	0.082	0.165	21.83
Expanded	Audio	Expanded Audio CNN - Tuned Threshold	Global threshold=0.2	0.382	0.041	0.379	3.30
Expanded	Hybrid	Expanded Hybrid Global Threshold	S=0.1, A=0.9, T=0.2	0.391	0.056	0.387	4.72
Expanded	Hybrid	Expanded Hybrid Per-Label + Best Cap	Per-label + cap=3	0.376	0.045	0.377	3.00

9. Formal FMA Test Evaluation: Notebook 49

Notebook 49 is the formal FMA test-split evaluation. This notebook matters because it uses corrected split selection and audio preprocessing. It processed 100 tracks, produced zero errors, and evaluated 150 labels.

Earlier empty-selection behaviour would have made this stage invalid, so the corrected record selection is part of the final result.

The Notebook 49 metrics table is best interpreted as internal test evidence. The Micro F1 and Samples F1 values show useful multi-label performance, while the low Macro F1 confirms that sparse labels remained difficult. This combination is consistent with the class imbalance seen earlier in the project.

Metric group	Precision	Recall	F1	Support
micro	0.374	0.566	0.450	295
macro	0.047	0.077	0.052	295
samples	0.389	0.645	0.445	100

Additional Notebook 49 metric	Value
Successful tracks	100
Error tracks	0
Stage 1 threshold	0.2
Hamming loss	0.0272
Jaccard samples	0.301
Low-confidence cases	12
Top-1 hit-any	72.0%
Top-3 hit-any	98.0%
Top-5 hit-any	99.0%

9.1 Per-Label Findings

Label	Precision	Recall	F1	Support
genre_8	1.000	1.000	1.000	1
genre_38	0.580	1.000	0.734	58
genre_47	1.000	0.500	0.667	2
genre_12	0.467	0.946	0.625	37
genre_32	0.536	0.600	0.566	25
genre_17	0.389	0.875	0.538	8
genre_15	0.367	0.786	0.500	28
genre_103	0.500	0.429	0.462	7
genre_250	0.333	0.500	0.400	6
genre_10	0.263	0.556	0.357	9

The best per-label results came from broad or better-supported genres. Macro F1 stayed much lower than Micro F1 because many labels had few examples. This is a common problem in genre work, but it is still important to state clearly. The report therefore uses per-label results and Top-k hit rates. A single accuracy score would hide too much.

9.2 Why Top-k Results Matter

The Top-k results are not included to make weak results look better. They match the problem. In music tagging, a useful system often needs to place a good candidate near the top, then let a reviewer or catalogue rule make the final choice. This is especially true when labels overlap or when the expected label is a broad family such as Rock, Electronic, or Experimental.

Notebook 49 shows this clearly. Top-1 hit-any was 72.0%, which is useful but not perfect. Top-3 and Top-5 were much stronger at 98.0% and 99.0%. That gap explains the final position of the project: the model is stronger as a ranked suggestion engine than as a single-label classifier.

The main technical meaning of the Notebook 49 results is the gap between Top-1 and Top-5. Top-1 hit-any of 72.0% shows that exact first-choice prediction was useful but still imperfect. Top-3 at 98.0% and Top-5 at 99.0% show that the model usually placed a correct or acceptable genre very near the top. In a music library workflow, that difference matters because the reviewer does not need the model to be the final authority. The reviewer needs a short and credible shortlist.

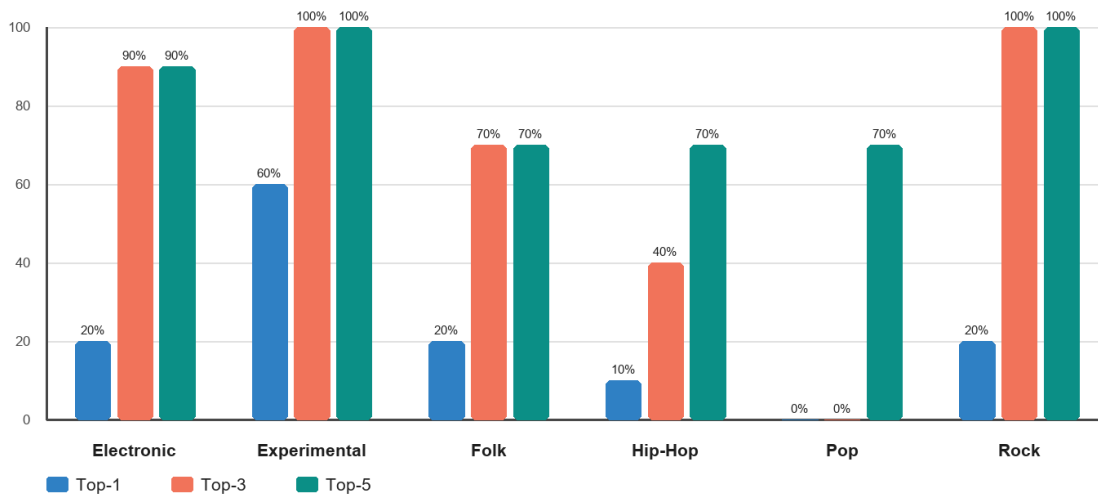
The low Macro F1 remains important and should not be ignored. It shows that the rare end of the label space did not receive the same quality of prediction as the common labels. This is why the final paper separates the 150 direct labels from the rare-tail fallback labels. The model can support discovery and review, but rare subgenre claims need caution.

10. External Audio Batch Evaluation: Notebook 50

Notebook 50 tests the model on external open-licence songs. The batch contains 60 tracks from Internet Archive Netlabels: 10 each for Rock, Electronic, Experimental, Folk, Hip-Hop, and Pop. The manifest records the song, artist, source URL, licence URL, target genre, accepted genre names, and accepted FMA label IDs. This keeps the test auditable and avoids commercial copyrighted downloads.

The Notebook 50 table should be read differently from the FMA test table. It measures transfer to external songs, not only performance on records drawn from the same project dataset. Lower Top-1 performance was therefore expected, but the Top-3 and Top-5 results still provide useful evidence for a ranked recommendation use case.

External Top-k Hit Rates by Target Genre



External testing shows uneven genre transfer, with Top-3 and Top-5 more useful than Top-1 alone.

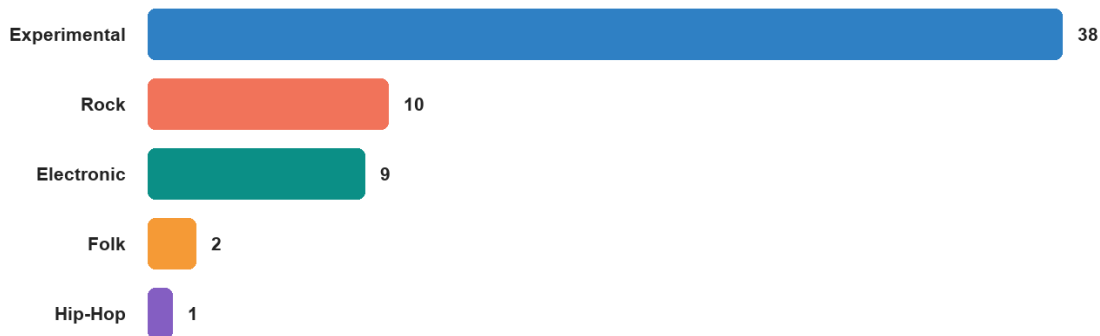
Figure 5. External Top-k hit rates by target genre.

Figure 5 should be read as evidence of uneven external transfer. Some target genres were easier to recover than others, and the value of Top-k results was clearer than the value of Top-1 alone. This supports the decision to present external testing as deployment-style evidence rather than final production validation.

Expected genre	Songs	Top-1	Top-3	Top-5	Stable hit	Low confidence
Electronic	10	20.0%	90.0%	90.0%	90.0%	50.0%
Experimental	10	60.0%	100.0%	100.0%	100.0%	50.0%
Folk	10	20.0%	70.0%	70.0%	70.0%	60.0%
Hip-Hop	10	10.0%	40.0%	70.0%	40.0%	70.0%
Pop	10	0.0%	0.0%	70.0%	0.0%	30.0%
Rock	10	20.0%	100.0%	100.0%	100.0%	30.0%

External overall metric	Value
Songs attempted	60
Successful songs	60
Errors	0
Top-1 hit rate	21.7%
Top-3 hit rate	66.7%
Top-5 hit rate	83.3%
Stable prediction hit rate	66.7%
Average main score	0.512
Low-confidence rate	48.3%

Main Prediction Distribution for External Songs



The concentration around broad labels explains why external outputs should be reviewed as ranked suggestions with confidence flags.

Figure 6. Main prediction distribution for external songs.

Figure 6 gives context for the external error pattern. The concentration of broad predictions suggests that the model sometimes favours general FMA labels when external audio does not match the training distribution closely.

The external test shows the practical weakness most clearly. Top-1 accuracy is low. Top-3 and Top-5 results are much more useful. That pattern fits the task. External songs do not always sound like the FMA training tracks, and accepted genres may be broader than a single label. The result is still useful if the model is treated as a ranking tool rather than a final judge.

10.1 Reading the External Results Carefully

The external batch should be read as a stress test, not as a clean copy of the FMA test split. The songs came from open-licence external sources, and their genre labels were mapped to acceptable FMA labels through a manifest. That mapping is useful, but it is still less controlled than the original dataset labels.

The main weakness was broad-label attraction. Experimental appeared often as a main prediction. This does not make the whole system unusable, but it shows that confidence thresholds and genre-family constraints would be needed before using the model for catalogue tagging.

External evaluation was included because a model can look strong on a held-out dataset split and still behave differently on songs collected elsewhere. The external batch used legal open-licence songs and accepted-genre mappings so that the test could be audited. This made the test more realistic, but also less controlled than FMA. Audio quality, genre naming, mixing style, and metadata expectations may all differ from the training environment.

The external results therefore act as a boundary check. Top-1 of 21.7% is not strong enough for automatic tagging. Top-3 of 63.3% and Top-5 of 83.3% are more useful because they show that the correct family often appears in the ranked list. The practical reading is that a demo should show several candidates with scores, not only one label. This is also why a low-confidence flag belongs in the final presentation.

An alternative approach would have been to train or fine-tune on the external songs, but that would have weakened the purpose of the test. The point of Notebook 50 was to see how the existing FMA-trained system transferred to new audio. A larger future study could add a separate external validation set, but the current result is still useful because it exposes the gap between internal test performance and deployment-style behaviour.

11. Limitations, Risks, and Honest Interpretation

Limitation	Evidence	Project Interpretation
Class imbalance	Macro F1 is much lower than Micro F1.	The model works better on common labels than rare labels.
External domain shift	External Top-1 is 21.7%, while Top-5 is 83.3%.	The model works better for ranked suggestions than exact one-label tags.
Experimental bias	38 of 60 external main predictions were Experimental.	The model often favours broad FMA labels on external audio.
Rare-tail weakness	Stage 2 rare-tail results remain limited.	Rare-tail output should be treated as supportive, not definitive.
External metadata gap	External songs do not have trusted FMA structured metadata rows.	Notebook 50 correctly uses audio-only inference for external songs.

The work should not be presented as a finished commercial tagging product. It is a working academic pipeline with clear evidence and clear limits. Results indicated that likely genres can be ranked usefully, but rare labels, external calibration, and broad-label bias remain open problems. This is why the final interpretation is careful rather than promotional.

Several limitations came from the structure of the data rather than from one isolated coding issue. FMA labels are uneven, some genres have very few examples, and broad parent genres can dominate predictions. These issues make rare-tail performance difficult and explain why the fallback stage is presented cautiously.

A second limitation is the external test setting. External songs are useful because they resemble deployment more closely than a held-out FMA split, but they also introduce uncertainty in genre mapping, audio quality, and musical style. For that reason, Notebook 50 is treated as deployment-style evidence, not as a direct replacement for the formal FMA test.

11.1 Lessons Learned from Failed or Weak Runs

Several weak runs were useful because they showed what the evaluation could not support. The empty test-selection output in the final testing phase was one example. It did not show that the model had failed; it showed

that the evaluation setup was wrong. Correcting that issue was necessary before any metric could be interpreted.

A second lesson was that longer audio does not automatically improve a model unless the inference method matches the training design. The audio model was trained around 15-second inputs, so Notebook 50 used windows across a full song rather than forcing a complete track into one prediction. This choice kept inference consistent with training while still allowing the whole song to influence the final ranked result.

A third lesson concerned scope. It was tempting to describe the final taxonomy as a full 163-genre classifier. The evidence did not support that wording. The more accurate description is a 163-genre pipeline with 150 direct candidate labels, 10 rare-tail fallback labels, and 3 inventory-only labels. That distinction may look small, but it is central to the credibility of the project.

The weaker results were also useful because they shaped the next stage of work. Class balancing, calibration, hierarchy-aware loss functions, genre-family constraints, and wider external testing are not optional polish. They are the next technical steps required before the system could be used as a stronger catalogue tool.

The external-song work also changed the interpretation. At first, it was tempting to expect full-song windowing to give a clear improvement because more of the track was being used. The results were more mixed. Windowing made the test more realistic, but it also exposed the limits of a model trained around 15-second FMA-style inputs. That trade-off is part of the final conclusion.

12. Reproducibility, Artefacts, and Submission Package

The notebooks, models, processed data, and output folders should be submitted with the paper. The report explains the decisions. The artefacts show the actual implementation. This matters because a reader can trace the final numbers back to the notebooks and output files.

Artefact group	Location	Purpose
Notebooks	E:\SCHOOL\Masters\Capstone_FMA_Project\notebook	Full build record from Notebook 01 to Notebook 50.
Models	E:\SCHOOL\Masters\Capstone_FMA_Project\models	Saved trained models, including the final audio model.
Processed data	E:\SCHOOL\Masters\Capstone_FMA_Project\data\processed	Prepared label matrices, summaries, thresholds, label arrays, and comparison tables.
Notebook 49 outputs	outputs\notebook49_fma_test_batch_evaluation	Formal FMA processed test evaluation outputs.
Notebook 50 outputs	outputs\notebook50_external_song_batch_windowed	External song predictions, manifest evaluation, and Top-k outputs.
External manifest	data\external_audio_batch\external_audio_manifest.xlsx	Audit file for external songs, sources, licences, and acceptable labels.
Graded proposal	documentation\Capstone_Project_Proposal_Graded.pdf	Approved proposal included as a supporting submission document.
Documentation	documentation	Final technical document and supporting documentation files.

12.1 Recommended Run Order for Review

The review order below is not meant to imply that every notebook carries equal weight. Earlier notebooks explain how the data and models were built. The final judgement should rely most heavily on the corrected final evaluation notebooks and the saved result files.

- Review Notebooks 23-24 for full genre inventory and multi-label preparation.
- Review Notebooks 34-38 for the expanded Candidate-150 audio and hybrid benchmark.
- Review Notebooks 39-47 for the corrected full 163-genre and rare-tail strategy.
- Run Notebook 49 to reproduce the final formal FMA processed test evaluation.
- Run Notebook 50 to reproduce the external open-licence audio batch evaluation.

12.2 What a Reviewer Should Check First

A reviewer should start with the final outputs rather than opening every notebook in order. The most useful files are the Notebook 49 summary JSON, the Notebook 49 top-k prediction table, the Notebook 50 batch summary JSON, and the Notebook 50 song summary table. These files show the final behaviour of the model without requiring a full rerun.

This review order also reflects the practical limits of rerunning the whole project. Audio models, external files, and notebook environments can be time-consuming to reproduce. The saved outputs and exported notebooks make the submission easier to audit without requiring every experiment to be rerun from scratch.

The notebook PDFs in the software artefacts folder are still important. They show the code and the output state used for submission. They also make the project easier to inspect on a machine that is not set up to rerun TensorFlow, librosa, and the full audio pipeline.

13. Conclusion and Future Work

The completed capstone provides a working, auditable music genre prediction pipeline built from FMA metadata, engineered features, raw audio, model outputs, and external-song testing. The most defensible conclusion is not that genre classification has been solved. The evidence supports a narrower and more useful claim: within this project, audio-based and hybrid modelling can rank likely genre candidates well enough to support review.

Notebook 49 gives the strongest internal evidence. The system completed 100 FMA test predictions with no errors and achieved strong Top-k performance. Notebook 50 gives the more cautious external evidence. It showed that the model still placed many acceptable genres in the Top-5, but that Top-1 external prediction remained limited. Taken together, these findings support a demonstration focused on ranked suggestions, confidence scores, and human review.

Future work should focus on calibration, rare-label learning, broader external testing, and a small serving interface for upload-based prediction. Those steps would move the capstone from a strong academic prototype

toward a more operational music analytics tool. The value of the current work is that it leaves that path clear, with the strengths and limits documented from the actual notebook evidence.

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Appendix A: Complete Notebook Artefact Map

Notebook	Project phase	Main purpose
Notebook 1: Metadata understanding	Foundations and proof of concept	Metadata understanding
Notebook 2: Structured model features	Foundations and proof of concept	Structured model features
Notebook 3: Echonest experiment	Foundations and proof of concept	Echonest experiment
Notebook 4: Mongodb pipeline	Foundations and proof of concept	Mongodb pipeline
Notebook 5: Structured model medium	Foundations and proof of concept	Structured model medium
Notebook 6: Large subset analysis and structured model	Foundations and proof of concept	Large subset analysis and structured model
Notebook 7: Audio model	Foundations and proof of concept	Audio model
Notebook 8: Improved audio model from large	Foundations and proof of concept	Improved audio model from large
Notebook 9: Structured model for audio hybrid	Foundations and proof of concept	Structured model for audio hybrid
Notebook 10: Hybrid fusion model	Foundations and proof of concept	Hybrid fusion model
Notebook 11: Final prediction pipeline	Foundations and proof of concept	Final prediction pipeline

Notebook	Project phase	Main purpose
Notebook 12: Expand 8genre pilot and retrain	Controlled genre expansion	Expand 8genre pilot and retrain
Notebook 13: Audio CNN expanded	Controlled genre expansion	Audio CNN expanded
Notebook 14: Structured model expanded	Controlled genre expansion	Structured model expanded
Notebook 15: Hybrid fusion expanded	Controlled genre expansion	Hybrid fusion expanded
Notebook 16: Build 10genre pilot	Controlled genre expansion	Build 10genre pilot
Notebook 17: Audio CNN 10genre	Controlled genre expansion	Audio CNN 10genre
Notebook 18: Structured model 10genre	Controlled genre expansion	Structured model 10genre
Notebook 19: Hybrid fusion 10genre	Controlled genre expansion	Hybrid fusion 10genre
Notebook 20: Final prediction pipeline 10genre	Controlled genre expansion	Final prediction pipeline 10genre
Notebook 21: Batch inference demo 10genre	Controlled genre expansion	Batch inference demo 10genre
Notebook 22: Final results summary and comparison	Controlled genre expansion	Final results summary and comparison
Notebook 23: Full genre inventory and multilabel prep	Candidate-150 multi-label system	Full genre inventory and multilabel prep
Notebook 24: Multilabel target matrix and split audit	Candidate-150 multi-label system	Multilabel target matrix and split audit
Notebook 25: Mongodb ingestion for multilabel phase	Candidate-150 multi-label system	Mongodb ingestion for multilabel phase
Notebook 26: Structured multilabel baseline Candidate-150	Candidate-150 multi-label system	Structured multilabel baseline Candidate-150
Notebook 27: Structured multilabel threshold tuning Candidate-150	Candidate-150 multi-label system	Structured multilabel threshold tuning Candidate-150
Notebook 28: Structured multilabel per label thresholds and topk decoding	Candidate-150 multi-label system	Structured multilabel per label thresholds and topk decoding
Notebook 29: Structured multilabel final upgrade Candidate-150	Candidate-150 multi-label system	Structured multilabel final upgrade Candidate-150
Notebook 30: Audio multilabel baseline Candidate-150 pilot	Candidate-150 multi-label system	Audio multilabel baseline Candidate-150 pilot
Notebook 31: Structured predictions for audio pilot Candidate-150	Candidate-150 multi-label system	Structured predictions for audio pilot Candidate-150
Notebook 32: Hybrid multilabel fusion Candidate-150 pilot	Candidate-150 multi-label system	Hybrid multilabel fusion Candidate-150 pilot

Notebook	Project phase	Main purpose
Notebook 33: Hybrid multilabel per label thresholds Candidate-150 pilot	Candidate-150 multi-label system	Hybrid multilabel per label thresholds Candidate-150 pilot
Notebook 34: Audio multilabel Candidate-150 expanded pilot	Candidate-150 multi-label system	Audio multilabel Candidate-150 expanded pilot
Notebook 35: Structured predictions for expanded audio Candidate-150	Candidate-150 multi-label system	Structured predictions for expanded audio Candidate-150
Notebook 36: Hybrid multilabel fusion Candidate-150 expanded	Candidate-150 multi-label system	Hybrid multilabel fusion Candidate-150 expanded
Notebook 37: Expanded hybrid multilabel per label thresholds and cap	Candidate-150 multi-label system	Expanded hybrid multilabel per label thresholds and cap
Notebook 38: Final Candidate-150 consolidation and report tables	Candidate-150 multi-label system	Final Candidate-150 consolidation and report tables
Notebook 39: Full 163 rare tail preparation and all genre strategy	Full 163-genre and rare-tail strategy	Full 163 rare tail preparation and all genre strategy
Notebook 40: Full 163 rare tail fallback router	Full 163-genre and rare-tail strategy	Full 163 rare tail fallback router
Notebook 41: Full 163 end to end inference pipeline	Full 163-genre and rare-tail strategy	Full 163 end to end inference pipeline
Notebook 42: Full 163 single file inference demo	Full 163-genre and rare-tail strategy	Full 163 single file inference demo
Notebook 43: Full 163 batch inference demo	Full 163-genre and rare-tail strategy	Full 163 batch inference demo
Notebook 44: Full 163 targeted rare tail batch demo	Full 163-genre and rare-tail strategy	Full 163 targeted rare tail batch demo
Notebook 45: Full 163 rare tail prototype reranker	Full 163-genre and rare-tail strategy	Full 163 rare tail prototype reranker
Notebook 46: Final full project consolidation	Full 163-genre and rare-tail strategy	Final full project consolidation
Notebook 47: Final audio inference pipeline	Full 163-genre and rare-tail strategy	Final audio inference pipeline
Notebook 47: Final audio inference pipeline patched multwindow safe stage2	Full 163-genre and rare-tail strategy	Final audio inference pipeline patched multwindow safe stage2
Notebook 48: Looped audio genre prediction with confidence	Final inference and evaluation	Looped audio genre prediction with confidence
Notebook 49: Final FMA test split batch evaluation	Final inference and evaluation	Final FMA test split batch evaluation
Notebook 50: External song batch windowed inference	Final inference and evaluation	External song batch windowed inference

Appendix B: Final Performance Snapshot

Area	Metric	Value	Interpretation
Candidate-150 benchmark	Best benchmark Micro F1	0.391	Best Candidate-150 benchmark.
Candidate-150 benchmark	Best conservative Micro F1	0.376	Cleaner optional output mode.
Full 163 random batch demo	Candidate hit-any rate	0.900	Batch demo result on a random sample.
Full 163 random batch demo	Average predicted candidate labels	4.33	Average number of Stage-1 direct predictions per sampled track
Full 163 random batch demo	Average rare-tail suggestions	0.467	Average Stage-2 suggestions per sampled track
Full 163 targeted rare-tail demo	Candidate hit-any rate	0.857	Stage-1 behaviour on rare-tail-containing tracks
Full 163 targeted rare-tail demo	Stage-2 suggestion coverage	0.657	How often Stage-2 surfaced a suggestion on targeted rare-tail tracks
Full 163 targeted rare-tail demo	Rare-tail hit-any rate	0.029	Pre-reranker Stage-2 success rate on targeted rare-tail tracks
Full 163 targeted rare-tail demo	Rare-tail exact top-1 hit rate	0.029	Pre-reranker top suggestion accuracy on targeted rare-tail tracks
Stage-2 reranker comparison	Baseline rare-tail hit-any rate	0.286	Baseline Stage-2 rare-tail test performance
Stage-2 reranker comparison	Best Stage-2 rare-tail hit-any rate	0.314	Final frozen Stage-2 rare-tail test performance

Appendix C: External Test Set Manifest Summary

Target genre	Songs	Primary acceptable labels
Rock	10	Rock, Punk, Lo-Fi, Indie-Rock, Garage, Psych-Rock, Industrial, Metal, Post-Rock and related labels.
Electronic	10	Electronic, Ambient Electronic, IDM, Glitch, Techno, Downtempo, Trip-Hop, House, Dubstep and related labels.
Experimental	10	Experimental, Avant-Garde, Noise, Electroacoustic, Improv, Drone, Sound Collage and related labels.
Folk	10	Folk, Singer-Songwriter, Psych-Folk, Freak-Folk and Free-Folk.
Hip-Hop	10	Hip-Hop, Hip-Hop Beats, Alternative Hip-Hop, Rap, Abstract Hip-Hop and Breakbeat.
Pop	10	Pop, Experimental Pop, Synth Pop and Power-Pop.

Appendix D: Source Materials Used

Source base: approved proposal, course outline, earlier project drafts, reference presentation, capstone notebooks, CSV and JSON outputs, project diagrams, and the Notebook 49 and Notebook 50 result files stored in the project folder.

The source materials were used to cross-check the project story against the files that were actually produced. The proposal and course outline were used for scope and assessment alignment. The notebooks and output files were used for the technical claims, especially the Notebook 49 and Notebook 50 results. Older drafts and diagrams were treated as background only where they matched the corrected 163-genre interpretation.